Velocity Loss as a Critical Variable Determining the Adaptations to Strength Training

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ABSTRACT

PAREJA-BLANCO, F., J. ALCAZAR, J. SÁNCHEZ-VALDEPEÑAS, P. J. CORNEJO-DAZA, F. PIQUERAS-SANCHIZ, R. MORA-VELA, M. SÁNCHEZ-MORENO, B. BACHERO-MENA, M. ORTEGA-BECERRA, and L. M. ALEGRE. Velocity Loss as a Critical Variable Determining the Adaptations to Strength Training. *Med. Sci. Sports Exerc.*, Vol. 52, No. 8, pp. 1752–1762, 2020. **Purpose:** This study aimed to compare the effects of four resistance training (RT) programs with different velocity loss (VL) thresholds: 0% (VL0), 10% (VL10), 20% (VL20), and 40% (VL40) on sprint and jump performance, muscle strength, neuromuscular, muscle hypertrophy, and architectural adaptations. **Methods:** Sixty-four young resistance-trained men were randomly assigned into four groups (VL0, VL10, VL20, and VL40) that differed in the VL allowed in each set. Subjects followed an RT program for 8 wk (two sessions per week) using the full-squat (SQ) exercise, with similar relative intensity (70%–85% 1-repetition maximum), number of sets (3), and interset recovery period (4 min). Before and after the RT program, the following tests were performed: 1) muscle hypertrophy and architecture of the vastus lateralis (V_{LA}), 2) tensiomyography, 3) 20-m running sprint, 4) vertical jump, 5) maximal voluntary isometric contraction in SQ, 6) progressive loading test in SQ, and 7) fatigue test. **Results:** No between-group differences existed for RT-induced gains in sprint, jump, and strength performance despite the differences in the total volume performed by each group. VL20 and VL40 showed significant increases ($P < 0.001$) in muscle hypertrophy (group–time interaction, $P = 0.06$). However, only VL40 exhibited a significant slowing ($P < 0.001$) of the delay time in the V_{LA} muscle (group–time interaction, $P = 0.05$). Moreover, VL40 showed a significant decrease in the early rate of force development ($P = 0.04$). **Conclusions:** Higher VL thresholds (i.e., VL20 and VL40) maximized hypertrophic adaptations, although an excessive VL during the set (i.e., VL40) may also induce negative neuromuscular adaptations. Therefore, moderate VL thresholds should be chosen to maximize strength adaptations and to prevent negative neuromuscular adaptations. **Key Words:** RESISTANCE TRAINING, DOSE–RESPONSE, NEUROMUSCULAR ADAPTATIONS, ARCHITECTURAL ADAPTATIONS, TENSIOMYOGRAPHY, MUSCLE HYPERTROPHY

Resistance training (RT) is recognized as an effective method for improving overall fitness (1). In this regard, the manipulation of training variables such as relative intensity (percent 1-repetition maximum (%1RM)), volume, exercise type and order, rest duration, and repetition velocity is a critical factor in order to induce neural and morphological adaptations (2,3). Notably, the dose–response relationship between training volume and athletic performance adaptations has been suggested to exhibit an inverted U-shaped curve (4–6). A recent study by Barbalho et al. (7) reported the existence of a plateau in strength gains and muscle hypertrophy after a threshold volume was reached (5–10 sets per muscle group per session). Further increases in training volume did not yield any gains and even led to declines in performance. However, most investigations have focused on the dose–response relationship between RT volume and RT-induced adaptations by manipulating the number of sets (2,7–9), whereas other alternatives remain unexplored.

Among the variables that can be manipulated to configure training volume, the actual number of repetitions performed in a set in relation to the maximal number of repetitions that

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can be completed (i.e., "level of effort") (10) seems to be another variable that should be considered when designing an RT program (10–12). Exercise sets are traditionally completed by performing every repetition in succession until the desired number of repetitions is reached (1,3). However, because the number of repetitions that can be completed against a given %1RM has shown a large variability between individuals (13,14), this traditional approach can lead to different levels of effort among individuals performing the same number of repetitions per set. It has been shown that the level of fatigue increases gradually as the level of effort during the set increases (10). In this regard, the gradual decrease in lifting velocity across repetitions performed within a set can be interpreted as evidence of impaired neuromuscular function, and its assessment may provide a simple and objective means of quantifying levels of fatigue (10). Moreover, there is a strong positive relationship between the percentage of velocity loss (VL) induced during the set and the percentage of repetitions that can be completed before muscle failure (R^2 = 0.93–0.97) (14). Thus, the monitoring of repetition velocity and of relative VL during training provides real-time feedback on the level of effort exerted in the set, with considerable accuracy. This novel strategy allows for the application of a more homogeneous stimulus across individuals, which might be useful in protocols aimed at studying adaptations to different levels of fatigue during RT sessions, because a standardized state of fatigue is induced.

A recent study (15) compared the effects of two squat training programs that only differed in the relative VL reached in each set: 20% (VL20), leading to the completion of approximately 50% of the possible repetitions per set, versus 40% (VL40), leading to the completion of almost 100% of the possible repetitions per set (i.e., very close to muscle failure). The VL20 group achieved similar or even superior strength gains, especially in high-velocity actions (e.g., vertical jump), whereas the VL40 group experienced a higher hypertrophic response and a IIX-to-IIA shift in single muscle fiber phenotype, whereas the VL20 group did not display reductions in the IIX fiber-type pool (15). Another study comparing a relative VL of 15% (VL15) versus 30% (VL30) reported similar results, pointing out the superiority of VL15 in terms of RT-induced strength and jumping gains (16). These findings may also suggest the existence of a volume plateau, beyond which further volume increases do not induce greater strength gains and can even induce suboptimal adaptations. However, further studies considering a wider spectrum of VL thresholds are needed. Therefore, the main goal of the present study was to compare the effects of four RT programs with different VL thresholds (0%, 10%, 20%, and 40%) on sprint and jump performance, muscle strength, neuromuscular, muscle hypertrophy, and architectural adaptations.

MATERIALS AND METHODS

Experimental Design

A longitudinal research design was used to compare the effects of four RT programs that only differed in the magnitude of the VL induced during the set: 0% (VL0) versus 10% (VL10) versus 20% (VL20) versus 40% (VL40). Subjects performed an 8-wk progressive lower limb RT program comprising only the full squat (SQ) exercise. The four groups trained at the same %1RM in each session. Sessions were performed in a research laboratory under the direct supervision of the investigators, at the same time of the day (±1 h) and the same environmental conditions (20°C and 60% humidity) for each subject. Subjects were requested not to perform any other type of strenuous physical activity during the study period. All groups were measured on two occasions: 72 h before (pretraining) and 72 h after (posttraining) the 8-wk training intervention. A battery of tests was performed in two testing sessions (separated by 48 h). The first testing session consisted of 1) ultrasound measurements of the vastus lateralis (V_{LA}) muscle and 2) tensiomyography (TMG) of the V_{LA} and vastus medialis (V_{ME}) muscles. On the second testing session, a battery of fitness tests was performed as follows: 1) 20-m running sprint, 2) vertical jump, 3) maximal voluntary isometric contraction (MVIC) in SQ exercise, 4) progressive loading test in SQ, and 5) fatigue test in SQ. Training compliance was 100% of all sessions for the subjects who completed the intervention.

Subjects

Sixty-four resistance trained and physically active men (age, 24.1 ± 4.3 yr; height, 1.75 ± 0.06 m; body mass, 75.5 ± 9.7 kg) volunteered to take part in this study. All subjects had a training background ranging from 1.5 to 4 yr (1–3 sessions per week) and were accustomed to performing the SQ exercise using the correct technique. Subjects were randomly assigned to one of four groups, which differed only in the magnitude of VL allowed in each training set. A participant flow diagram is presented in Figure 1. After being informed of the purpose and testing procedures, subjects signed a written informed consent form before participation. The present study was approved by the local research ethics committee, in accordance with the Declaration of Helsinki. No physical limitations, health problems, or musculoskeletal injuries that could affect testing were found after a medical examination. Subjects reported to be free from taking drugs, medications, or dietary supplements known to influence physical performance. Caffeine consumption was not permitted 12 h before the TMG trials.

Testing Procedures

Ultrasonography. Muscle cross-sectional area (CSA) and the architecture of the V_{LA} muscle were assessed using B-mode ultrasonography (MyLab 25; Esaote Biomedica, Genoa, Italy) with a 50-mm, 5- to 12-MHz linear array probe. Resting ultrasound images were collected with the participants lying in the supine position with the knees resting slightly flexed at 150° (180° = full knee extension). After 15 min in the described position, V_{LA} CSA and architecture were measured at 50% and 60%, respectively, of the distance between the superior border of the greater trochanter and the inferior

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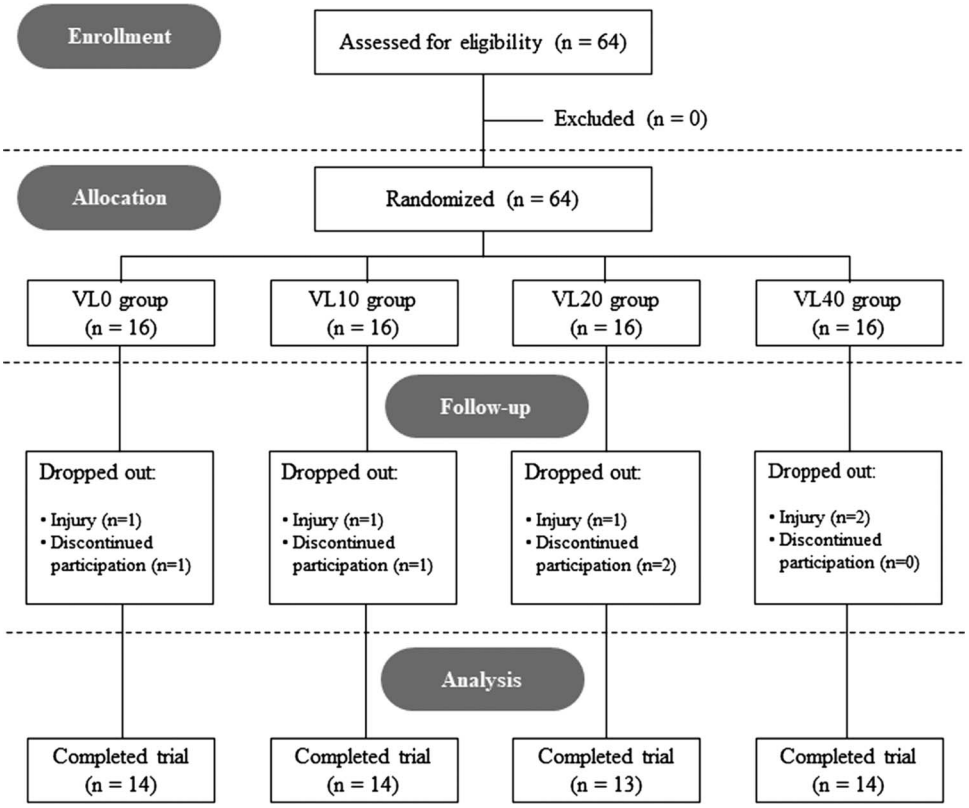

FIGURE 1—Participant flow diagram.

border of the lateral condyle of the right leg. The extended field of view mode was used in the case of muscle CSA, whereas the alignment of the transducer in the fascicular plane was required in the case of the muscle architecture. Adhesive gaskets were placed on the skin at the proximal and distal boundaries of the transverse path to be measured to facilitate the application of the gel and the rectilinear path of the ultrasound linear array probe. The panoramic image was taken from medial to lateral, passing carefully and at a constant velocity along the defined path, keeping the probe perpendicular to the surface of the skin and applying a minimal pressure not to compress the tissues that will be evaluated. Three images were recorded at each thigh length (i.e., 50% and 60%) and digitally analyzed (ImageJ 1.51j8; NIH, Bethesda, MD) by the same operator, who was blinded to subject allocation. Muscle CSA (in centimeters squared) was measured by surrounding the bounds (aponeuroses) of the V_{LA} muscle (Fig. 2A). V_{LA} pennation angle (PA) and fascicle length (FL) were measured from the visible portion of two fascicles within the same image and linear extrapolation of fibers and aponeuroses when a portion of the fascicle extended off the field of view (Fig. 2B) (17). Two ultrasound images were assessed for each variable, and when a coefficient of variation (CV) greater than 5% was found, the third image was analyzed. The average value from all the analyzed images was considered for further analysis. Consistency in measurement sites across testing days was achieved by recording probe positions onto a transparent acetate and using easily identifiable infiltrations of fatty and connective tissues as landmarks. Test–retest CV for V_{LA} CSA was 3.8%, and CVs for muscle architecture measures using these procedures have been previously reported: 2.7% for FL and 3.6% for PA (18).

Tensiomyography. The V_{LA} and V_{ME} muscles' contractile properties were assessed by TMG (TMG-100 System electrostimulator; TMG-BMC, Ljubljana, Slovenia) as their response to an electrically evoked contraction. The electric stimulus was induced through two self-adhesive electrodes (interelectrode distance, 5 cm; Dura-Stick® premium; Cefar-Compex, Hanover, Germany), and the muscle mechanical response was measured with a digital Dc-Dc transducer Trans-TekR (GK 40, Ljubljana, Slovenia) placed perpendicular to the muscle belly and equidistant from the self-adhesive electrodes at a distance of 25–30 mm. Participants remained lying in the supine position for 10 min before starting the TMG measurements data acquisition, and the V_{LA} and V_{ME} were marked according to the SENIAM indications and location (19). Measurements were taken with the athletes in the supine position and the knee joint fixed at an angle of ~140° using a wedge cushion. Electrical stimulation was applied with a pulse duration of 1 ms and an initial current amplitude of 40 mA, which was progressively increased in 10-mA steps up to the stimulator's maximal output (100 mA). A 10-s rest period was allowed between each electrical stimulus to avoid fatigue or posttetanic activation (20). The variables assessed in this study were the maximal radial displacement of the muscle belly (Dm), contraction time (Tc), and delay time (Td). Dm was defined as

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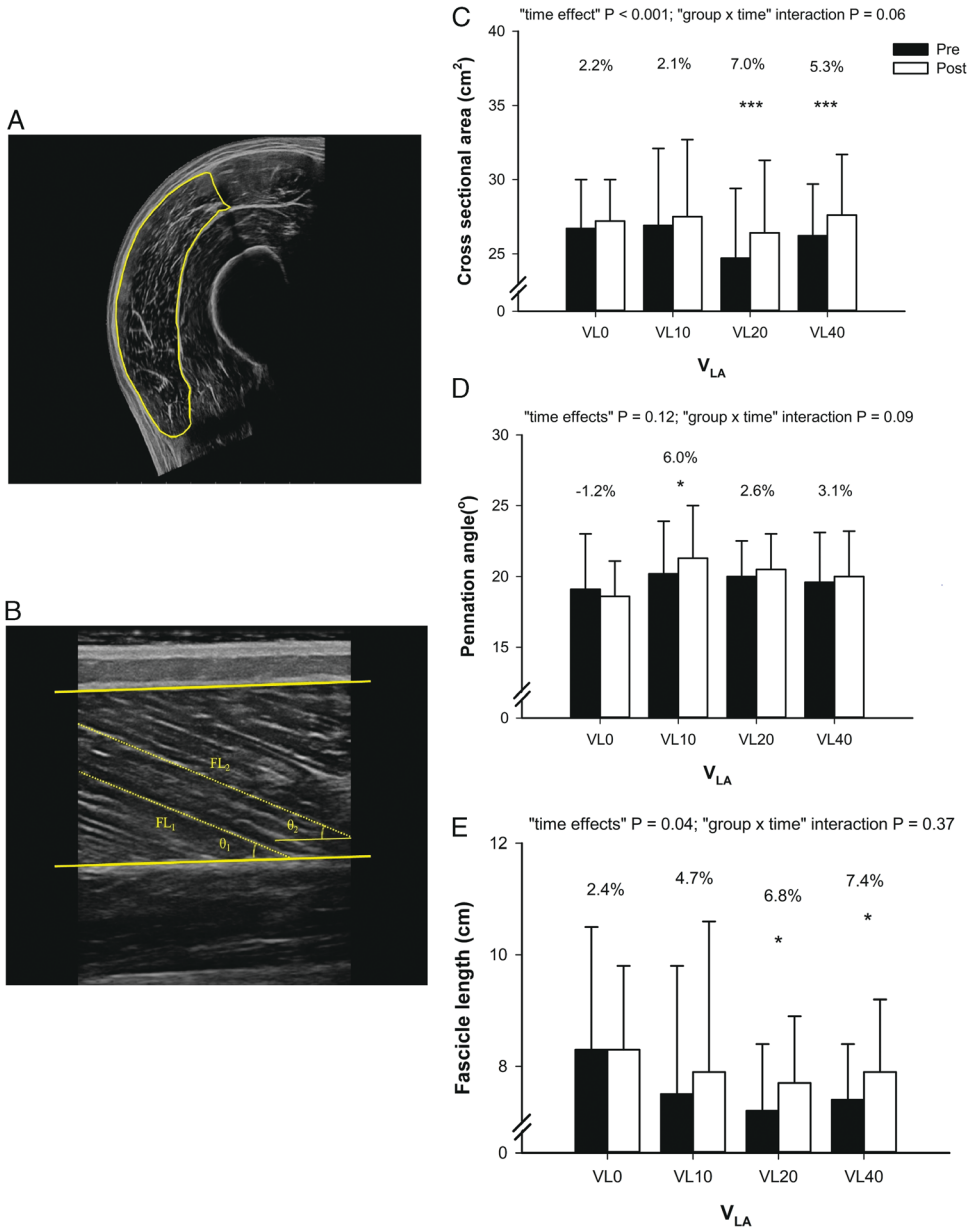

FIGURE 2—Ultrasound images obtained from the V_{LA} of a standard subject. A, The CSA is surrounded by a thick line. B, Thick lines indicate the superior and inferior aponeuroses of the V_{LA} muscle, dashed lines indicate the visible portions of muscle fascicles, and arched lines indicate angles formed between the visible muscle fascicles and the inferior aponeurosis. Nonvisible portions of muscle fascicles were estimated using linear extrapolation of fibers and aponeuroses in order to estimate FL. Changes produced from pretraining to posttraining for each group in CSA (C), PA (D), and FL (E) of V_{LA} obtained from ultrasound images. Data are mean ± SD, *N* = 55. VL0: group that trained with a mean VL of 0% in each set (*n* = 14); VL10: group that trained with a mean VL of 10% in each set (*n* = 14); VL20: group that trained with a mean VL of 20% in each set (*n* = 13); VL40: group that trained with a mean VL of 40% in each set (*n* = 14). θ, PA; Δ (%), relative change from pretraining to posttraining. Intra-group significant differences from pretraining to posttraining: **P* < 0.05, ****P* < 0.001.

the peak amplitude in the displacement–time curve of the tensiomyographical twitch response; Tc was obtained by determining the time interval from 10% to 90% of Dm; and Td was defined as the time between the electrical stimulus and 10% of Dm (21). All measurements were carried out by the same expert technician, and only the curve with the highest Dm value was considered for further analysis. Test–retest reliability for the V_{ME} and V_{LA} muscles, respectively, was as follows: CV = 5.5% and 5.0% for Dm, CV = 2.6% and 3.4% for Tc, and CV = 3.1% and 3.0% for Td.

Running sprint and vertical jump tests. Subjects performed two maximal 20-m sprints on an indoor running track, separated by a 3-min rest between each sprint. Sprint times were measured using photocells (Witty; Microgate, Bolzano, Italy). Photocell timing gates were placed at 0, 10, and 20 m so that the times to cover the 0- to 10-m (T0–10), 10- to 20-m (T10–20), and 0- to 20-m (T0–20) distances could be determined. A standing start with the lead-off foot placed 1-m behind the first timing gate was used. The warm-up protocol consisted of four 20-m running accelerations at 80%, 85%, 90%, 95% of

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perceived effort, and one 10-m sprint at 100% effort with 1-min rest periods between them. The test–retest reliability (CV values) ranged from 0.7% to 1.4%.

In the countermovement jump (CMJ) test, subjects performed five maximal CMJs with 20-s rests between each jump. Before CMJ assessment, participants warmed up by performing two sets of 10 squats without external load, 5 submaximal CMJs, and 3 maximal CMJs. CMJ height was calculated from flight time values determined using an infrared timing system (OptojumpNext; Microgate, Bolzano, Italy). After discarding the highest and lowest CMJ heights, the resulting average was kept for further analysis. The CV was 1.6%.

MVIC test. Maximal isometric force (MIF) and maximal rate of force development (RFDmax) were measured during an MVIC in the SQ exercise with the participants standing with their knees flexed at 90° (180° = full extension). This test was performed on a Smith machine with height-adjustable movable supports, instrumented with an 80 × 80-cm dynamometric platform (FP-500; Ergotech, Murcia, Spain). The subjects were instructed to push with their legs on the force platform as fast and hard as possible after the cue "ready, set, go!" during two 5-s trials separated by 1 min of rest. External forces were collected at a sampling rate of 1000 Hz and processed with specific software (T-Force System; Ergotech). RFDmax was calculated as the maximal slope in the force–time curve in 20-ms time intervals. Moreover, the average tangential slope of the force–time curve obtained over different time intervals (50, 100, and 150 ms from the onset of force production, $RFD_{0–50}$, $RFD_{0–100}$, and $RFD_{0–150}$, respectively) was calculated. The average value of each variable in the two attempts was recorded for further analysis. Test–retest reliability values (CV values) for the MIF and RFD values, respectively, was 4.3% and from 12.8% to 24.3%.

EMG signal acquisition. After skin preparation, surface EMG electrodes were placed over the V_{ME} and V_{LA} muscles of the right leg following SENIAM recommendations (19). Consistency in measurement sites across testing days was achieved by recording electrode positions onto a transparent acetate and using different anatomic references and skin moles as landmarks. EMG signals were recorded continuously during MVIC testing using a parallel-bar, bipolar surface electromyographic sensor Trigno™ wireless EMG system, with an interelectrode distance of 10 mm, common mode rejection ratio >80 dB, and bandwidth filter between 20 and 450 Hz ± 10% (Delsys Inc, Natick, MA). Baseline noise was <5 μV peak-to-peak, and sampling rate was 1926 Hz. The raw data from the EMG were stored in digital format using EMG works Acquisition software (Delsys Inc). Root mean square (RMS) values were calculated to identify changes in muscle excitation levels. The highest RMS during each MVIC was calculated using a moving window of 500 ms with an overlap of 490 ms. V_{ME} and V_{LA} muscle excitation values were averaged, and the average of the two MVICs was calculated for further analysis. The CV for the RMS signal was 9.3%.

Progressive loading test. Individual load–velocity relationship and 1RM load in the SQ exercise were determined using a progressive loading test in a Smith machine (Multipower Fitness Line; Peroga, Murcia, Spain) with no counterweight mechanism. The SQ was performed with subjects starting from the upright position with the knees and hips fully extended, parallel feet and stance approximately shoulder-width apart, and the barbell resting across the back at the level of the acromion. Each subject descended in a continuous motion as low as possible (~35°–40° knee flexion), then immediately reversed motion and raised back to the upright position. Unlike the eccentric phase that was performed at a controlled mean velocity (~0.50–0.65 $m \cdot s^{-1}$), subjects were required to always execute the concentric phase at maximal intended velocity. Range of movement and velocity values of all repetitions were recorded at 1000 Hz with a linear velocity transducer (T-Force System; Ergotech), whose reliability has been reported elsewhere (10). First, the participants warmed up by performing six repetitions with a 20-kg load. The initial load was set at 20 kg and was progressively increased in 10-kg increments until the mean propulsive velocity (MPV) was ≤0.50 $m \cdot s^{-1}$. Then, the load was increased with smaller increments (2.5–5.0 kg) for better adjustment. A total of 8.5 ± 1.8 increasing loads were used for each subject. Three repetitions were executed for light loads (≥1.00 $m \cdot s^{-1}$), two for medium loads (1.00–0.80 $m \cdot s^{-1}$), and one for the heaviest loads (≤0.80 $m \cdot s^{-1}$). Interset recoveries ranged from 3 min (light loads) to 5 min (heavy loads). Only the best repetition (i.e., highest MPV) with each load was considered for subsequent analysis. The propulsive phase corresponds to the portion of the concentric action during which the measured acceleration is greater than acceleration due to gravity (−9.81 $m \cdot s^{-2}$) (22).

In addition to the individual load–velocity relationship and the corresponding 1RM load, we reported the following: a) average MPV attained against all absolute loads common to pretraining and posttraining (AV), b) average MPV attained against absolute loads that were moved faster than 1 $m \cdot s^{-1}$ at pretraining (AV > 1), and c) average MPV attained against absolute loads that were moved slower than 1 $m \cdot s^{-1}$ at pretraining (AV < 1). These variables were analyzed to examine the potential adaptations achieved in different areas of the load–velocity relationship.

Fatigue test. Five minutes after the SQ progressive loading test, the subjects were required to complete as many repetitions as possible with a load corresponding to 70% of 1RM until the MPV fell below 0.50 $m \cdot s^{-1}$. The test was repeated after the completion of the study using the same absolute load as in the baseline examination (i.e., 70% of baseline 1RM). The execution technique and devices used were those described for the progressive loading test. Subjects were encouraged to perform each repetition at maximal intended velocity. The total number of repetitions being performed was used for further analysis.

RT Program

The descriptive characteristics of the RT program are presented in Table 1. The four groups trained twice a week

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TABLE 1. Descriptive characteristics of the 8-wk velocity-based squat training program performed by the four experimental groups.

Scheduled								
	Training Sessions							
	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7	Session 8
Set × %1RM	3 × 70	3 × 70	3 × 70	3 × 70	3 × 70	3 × 75	3 × 75	3 × 75
	Session 9	Session 10	Session 11	Session 12	Session 13	Session 14	Session 15	Session 16
Set × %1RM	3 × 75	3 × 75	3 × 80	3 × 80	3 × 80	3 × 80	3 × 85	3 × 85

Actually Performed					
Training Group	Fastest MPV ($m \cdot s^{-1}$)	Slowest MPV ($m \cdot s^{-1}$)	MPV All Reps	Mean VL	Total Rep
VL0	0.77 ± 0.11	0.70 ± 0.12*,**,***	0.74 ± 0.11**,***	0 ± 0.0*,**,***	48.0 ± 0.0*,**,***
VL10	0.76 ± 0.08	0.61 ± 0.08**,***	0.71 ± 0.07***	10.6 ± 0.9**,***	143.6 ± 40.2***
VL20	0.73 ± 0.07	0.51 ± 0.06***	0.64 ± 0.06	20.3 ± 1.5***	168.5 ± 47.4***
VL40	0.75 ± 0.07	0.37 ± 0.05	0.58 ± 0.06	41.9 ± 1.4	305.6 ± 81.7
Training Group	**Average Rep per Set in All Sessions**	**Rep per Set with 70% 1RM**	**Rep per Set with 75% 1RM**	**Rep per Set with 80% 1RM**	**Rep per Set with 85% 1RM**
VL0	1.0 ± 0.0*,**,***	1.0*,**,***	1.0*,**,***	1.0*,**,***	1.0*,**,***
VL10	3.0 ± 0.8***	3.7 ± 0.9***	3.1 ± 0.8***	2.5 ± 0.9***	2.2 ± 0.9***
VL20	3.5 ± 1.0***	4.2 ± 1.2***	3.8 ± 1.2***	2.9 ± 0.8***	2.3 ± 0.8***
VL40	6.4 ± 1.7	7.6 ± 2.5	7.0 ± 2.3	5.3 ± 1.5	4.3 ± 1.5

Data are mean ± SD. Only one exercise (SQ) was used in training. VL0: group that trained with a mean VL of 0% in each set (n = 14); VL10: group that trained with a mean VL of 10% in each set (n = 14); VL20: group that trained with a mean VL of 20% in each set (n = 13); VL40: group that trained with a mean VL of 40% in each set (n = 14).
*Statistically significant differences with VL10 protocol: $P < 0.05$.
**Statistically significant differences with VL20 protocol: $P < 0.05$.
***Statistically significant differences with VL40 protocol: $P < 0.05$.
VL, magnitude of VL expressed as percent loss in mean repetition velocity from the fastest (usually first) to the slowest (last one) repetition of each set; Fastest MPV, average of the fastest repetitions measured in each session (this value represents the average intensity, %1RM, achieved during the training program); Slowest MPV, average of the slowest repetitions measured in each session; MPV all reps, average MPV attained during the entire training program; Mean VL, average VL attained during the entire training program; Total rep, total number of repetitions performed during the training program; Average rep per set in all sessions, average number of repetitions performed in each set; Rep per set with a given %1RM, average number of repetitions performed in each set with each of the loads used (70%, 75%, 80%, or 85% 1RM).

(training sessions being 48–72 h apart) for 8 wk with the same exercise (full SQ), relative intensity (increasing from 70% to 85% 1RM), number of sets (3), and interset recovery (4 min). The only difference between groups was the VL threshold allowed in each set. Hence, the VL0 group performed only one repetition per set (0% of VL), whereas the other groups (VL10, VL20, and VL40) stopped their sets when the corresponding VL threshold (10%, 20%, and 40% of VL, respectively) was exceeded. All repetitions for all participants during all sessions were recorded using a linear velocity transducer (T-Force System; Ergotech). Relative loads were determined from the individual load–velocity relationship obtained from the progressive loading test in the SQ exercise for each subject (R^2 = 0.99 ± 0.01). Therefore, for each training session, individual loads were adjusted to ensure the corresponding MPV matched (±0.03 $m \cdot s^{-1}$) the prescribed %1RM. In addition, subjects received immediate velocity feedback while being encouraged to perform each repetition at maximal intended velocity during each training session. A standardized warm-up preceded each training session: i) 5 min of jogging at a self-selected easy pace, ii) 2 × 10 repetitions of SQ (own body weight), iii) 4 × 6–6–4–3 repetitions of SQ (20 kg, 40%, 50%, and 60% of 1RM, respectively), and iv) 1 × 2 repetitions of SQ at 70% of 1RM (only in sessions 6–16) plus 1 × 1 repetition of SQ at 80% of 1RM (only in sessions 15–16).

Statistical Analysis

Values are reported as mean ± SD. Sample size was calculated (using GPower Version 3.1.9.2) introducing the following parameters: effect size index (1.05 for a 10% increase in 1RM based on previous data (15)); and α error probability (0.05) and power (0.95), which resulted in a sample size of 12 subjects per group. The CV was measured to observe the test–retest absolute reliability. Normality and homoscedasticity were verified with Shapiro–Wilk and Levene's tests, respectively. Data were analyzed using a 4 × 2 factorial ANOVA with Bonferroni's *post hoc* comparisons using one between-group factor (VL0 vs VL10 vs VL20 vs VL40) and one within-group factor (pretraining vs posttraining). Statistical significance was established at the $P \leq 0.05$ level. All statistical analyses were performed using SPSS software version 20.0 (SPSS Inc., Chicago, IL).

RESULTS

Attrition and compliance. Nine subjects dropped out during the course of the study; thus, the final sample was composed of 55 subjects as follows: VL0 (n = 14), VL10 (n = 14), VL20 (n = 13), and VL40 (n = 14) (Fig. 1). Compliance was 100% for all subjects. No significant differences between groups were observed at pretraining for any of the variables analyzed.

Muscle CSA and architecture. RT-induced adaptations in V_{LA} muscle CSA, PA, and FL are shown in Figure 2. A tendency to a group–time interaction was observed for V_{LA} muscle CSA (P = 0.06), with a significant time effect ($P < 0.001$). The VL20 and VL40 groups showed significant improvements in muscle CSA (+7.0% and +5.3%, respectively; both $P < 0.001$), whereas no significant changes were observed for VL0 and VL10 (Fig. 2C). No significant group–time interactions were observed for V_{LA} PA or FL, with a significant time effect for FL (P = 0.04). With regard to PA values, only the VL10 group exhibited significant changes in PA (+6.0%, P = 0.02; Fig. 2D), whereas FL

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TABLE 2. Changes in muscle contractile properties and squat performance from pretraining to posttraining for each group.

	VL0			VL10			VL20			VL40				
	Pre	Post	Δ (%)	Pre	Post	Δ (%)	Pre	Post	Δ (%)	Pre	Post	Δ (%)	*P* Value Time Effect	*P* Value Group–Time
TMG parameters														
V_{LA}-Tc (ms)	22.5 ± 3.4	21.8 ± 3.1	−2.8	21.3 ± 3.3	20.8 ± 3.0	−1.8	22.3 ± 2.6	21.1 ± 1.6	−4.6	21.6 ± 5.8	21.8 ± 4.1	3.4	0.19	0.70
V_{LA}-Td (ms)	22.9 ± 1.8	23.3 ± 2.1	1.8	22.6 ± 1.7	23.2 ± 1.3	3.0	23.6 ± 1.4	23.6 ± 1.4	0.3	22.1 ± 2.5	23.8 ± 2.6***	8.4	0.004	0.05
V_{LA}-Dm (mm)	5.37 ± 1.51	5.48 ± 1.76	2.3	6.01 ± 2.85	5.57 ± 1.94	−0.5	5.35 ± 1.33	4.95 ± 1.67	−7.7	4.93 ± 1.67	5.60 ± 2.05	17.5	0.94	0.24
V_{ME}-Tc (ms)	21.8 ± 2.1	22.0 ± 2.3	1.2	21.8 ± 2.0	21.5 ± 2.3	−1.8	20.8 ± 2.1	20.4 ± 1.4	−1.5	21.7 ± 2.6	21.4 ± 3.0	−1.3	0.24	0.58
V_{ME}-Td (ms)	21.4 ± 1.2	22.1 ± 1.4	3.4	21.8 ± 1.5	21.7 ± 1.0	−0.9	21.2 ± 1.6	21.1 ± 1.5	−0.5	21.3 ± 1.5	21.9 ± 1.5	3.0	0.20	0.30
V_{ME}-Dm (mm)	7.58 ± 1.75	6.81 ± 1.46*	−6.7	8.00 ± 2.01	7.14 ± 2.28*	−11.3	6.71 ± 1.73	6.09 ± 1.52	−6.9	6.95 ± 1.79	7.19 ± 1.60	5.8	0.008	0.12
Squat performance														
1RM (kg)	99.9 ± 19.1	112.9 ± 19.9***	13.7	96.7 ± 15.3	113.6 ± 15.6***	18.1	97.8 ± 23.2	110.9 ± 19.6***	14.9	94.5 ± 17.5	105.6 ± 18.5***	12.3	<0.001	0.27
AV ($m \cdot s^{-1}$)	0.99 ± 0.11	1.08 ± 0.12***	9.5	0.97 ± 0.09	1.11 ± 0.09***	14.3	0.97 ± 0.05	1.11 ± 0.08***	14.3	0.96 ± 0.09	1.06 ± 0.13***	10.5	<0.001	0.34
AV > 1 ($m \cdot s^{-1}$)	1.30 ± 0.09	1.35 ± 0.13*	4.5	1.29 ± 0.06	1.38 ± 0.09***	6.6	1.26 ± 0.06	1.36 ± 0.07***	8.1	1.26 ± 0.09	1.33 ± 0.08**	5.2	<0.001	0.52
AV < 1 ($m \cdot s^{-1}$)	0.67 ± 0.05	0.80 ± 0.11***	20.5	0.67 ± 0.05	0.85 ± 0.09***	27.3	0.68 ± 0.03	0.86 ± 0.09***	25.9	0.68 ± 0.05	0.80 ± 0.09***	17.7	<0.001	0.28
MNR (rep)	9.2 ± 2.5	14.2 ± 6.0**	70.9	10.0 ± 3.2	19.7 ± 7.9***	100.2	8.4 ± 2.3	17.1 ± 7.2***	103.8	8.4 ± 2.8	16.1 ± 7.5***	96.3	<0.001	0.19

Data are mean ± SD, $N = 55$. VL0: group that trained with a mean VL of 0% in each set ($n = 14$); VL10: group that trained with a mean VL of 10% in each set ($n = 14$); VL20: group that trained with a mean VL of 20% in each set ($n = 13$); VL40: group that trained with a mean VL of 40% in each set ($n = 14$).

*Intragroup significant differences from pretraining to posttraining: $P < 0.05$.

**Intragroup significant differences from pretraining to posttraining: $P < 0.01$.

***Intragroup significant differences from pretraining to posttraining: $P < 0.001$.

1RM: one-repetition maximal in full squat exercise; AV: average MPV attained against all absolute loads common to Pre- and Post-training; AV < 1: average MPV attained against absolute loads that were moved slower than 1 $m \cdot s^{-1}$ at Pre-training; AV > 1: average MPV attained against absolute loads that were moved faster than 1 $m \cdot s^{-1}$ at Pre-training; Dm: muscle displacement; MNR, maximal number of repetitions in the fatigue test; Tc: contraction time; Td: delay time; V_{LA}: vastus lateralis muscle; V_{ME}: vastus medialis muscle; (%), relative change from pretraining to posttraining.

increased significantly only in VL20 (6.8%, $P = 0.05$) and VL40 (7.4%, $P = 0.02$; Fig. 2E).

Tensiomyography. A significant group–time interaction ($P = 0.05$) and a main effect of time ($P = 0.004$) were observed for Td in the V_{LA} muscle (V_{LA}-Td), although only the VL40 group exhibited a significant increase ($P < 0.001$; Table 2). No significant group–time interactions were found for the rest of variables (Table 2). A significant main effect of time was also noted for V_{ME}-Dm ($P = 0.008$), with significant decreases being observed only in the VL0 ($P = 0.04$) and VL10 ($P = 0.02$) groups. No significant changes were observed in Tc, V_{LA}-Dm, or V_{ME}-Td.

Running sprint and vertical jump tests. No significant group–time interactions were found for any of the variables analyzed. A significant main effect of time ($P < 0.001$), coupled with significant increases in CMJ height, was found in VL0 (+5.6%), VL10 (+8.0%), VL20 (+5.4%), and VL40 (+6.1%; $P < 0.001$–0.01). A significant time effect was observed for T10–20 ($P < 0.001$). However, no significant changes were detected in any group for sprint-related variables, except for T10–20 performance, which improved in the VL10 group (−1.8%; $P = 0.002$).

MVIC- and EMG-derived outcomes. Figure 3 displays the main findings regarding MVIC- and EMG-derived outcomes. No significant group–time interactions were noted. Significant time effects were observed for MIF ($P < 0.001$), RFDmax ($P = 0.03$), and $RFD_{0–150}$ ($P = 0.04$). All groups showed significant improvements in MIF ($P < 0.001$–0.05; Fig. 3A). Moreover, VL0 showed significantly greater RFDmax values than did VL10 at posttraining ($P = 0.04$; Fig. 3B), and VL40 showed significant decreases in $RFD_{0–50}$ ($P = 0.04$; Fig. 3C). No significant intragroup changes were observed in $RFD_{0–100}$, $RFD_{0–150}$, and averaged V_{LA} and V_{ME} muscle excitation (RMS) values (Fig. 3D).

Individual load–velocity relationship, 1RM, and fatigue. RT-induced adaptations noted in dynamic SQ performance are shown in Table 2. No significant group–time interactions were found for any variable, although significant time effects were observed for all these variables (all $P < 0.001$). 1RM values increased significantly in VL0 (+13.7%), VL10 (+18.1%), VL20 (+14.9%), and VL40 (+12.3%), as did velocity-related outcomes obtained from the individual load–velocity relationship (AV, AV > 1, and AV < 1; all $P < 0.05$). Figure 4A shows the evolution of the RT-induced gains in 1RM values estimated from the individual load–velocity relationships. All four groups significantly improved with regard to the number of repetitions performed in the fatigue test ($P < 0.001$–0.01).

Training program. The main kinematic characteristics of the RT programs are reported in Table 1. No between-group differences were noted for the averaged fastest MPV during each session, whereas registered relative VL differed significantly among groups: 0% for VL0, 10.6% ± 0.9% for VL10, 20.3% ± 1.5% for VL20, and 41.9% ± 1.4% for VL40 (all $P < 0.05$). Thus, the average MPV attained during the training program decreased as the VL increased ($P < 0.05$). Moreover, the total number of repetitions and the average number of

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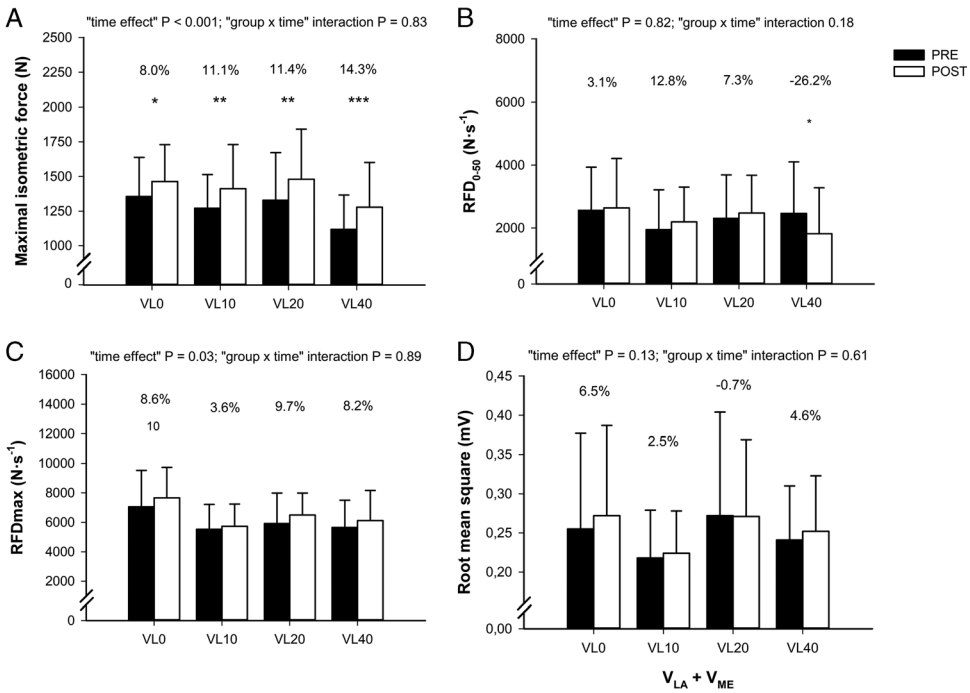

FIGURE 3—Changes produced from pretraining to posttraining for each group in MIF (A), RFDmax (B), rate of force development (C) from the onset of force production to 50 ms ($RFD_{0–50}$), and RMS (D) averaged from the V_{ME} and V_{LA}. Data are mean ± SD, *N* = 55. VL0: group that trained with a mean VL of 0% in each set (*n* = 14); VL10: group that trained with a mean VL of 10% in each set (*n* = 14); VL20: group that trained with a mean VL of 20% in each set (*n* = 13); VL40: group that trained with a mean VL of 40% in each set (*n* = 14). Δ (%), relative change from pretraining to posttraining. Intragroup significant differences from pretraining to posttraining: **P* < 0.05, ***P* < 0.01, ****P* < 0.001. Between-group significant differences (*P* < 0.05) with respect to VL10: 10.

repetitions per set (total and for each %1RM) increased as relative VL increased ($P < 0.05$). The total repetitions performed at different velocity ranges by each VL group are shown in Figure 4B.

DISCUSSION

The main findings derived from this study comparing four RT programs differing in VL threshold induced during the set were as follows: i) no between-group differences existed for RT-induced gains in sprint and jump performance, individual load–velocity relationship, MVIC, and neuromuscular adaptations, despite the considerable differences in the reps per set performed by each group (VL0, 1.0 ± 0.0; VL10, 3.0 ± 0.8; VL20, 3.5 ± 1.0; VL40, 6.4 ± 1.7), and ii) higher VL thresholds (i.e., VL20 and VL40) promoted an increased muscle hypertrophy response, although an excessive VL during the set may also induce negative neuromuscular adaptations (i.e., longer V_{LA}-Td and a decrease in $RFD_{0–50}$ only in VL40). Therefore, our data suggest that once an upper VL threshold is achieved a further increase in VL does not elicit additional strength gains and may even lead to negative neuromuscular adaptations.

All groups showed significantly increased SQ performance in terms of 1RM, MIF, AV, AV < 1, and AV > 1, although greater percentage changes were noted when applying moderate VL thresholds (i.e., VL10 and VL20). Thus, only one repetition per set (VL0) may not be enough to maximize strength gains, whereas a high VL threshold (VL40) does not provide further strength or velocity gains. A recent review concluded that a single set of 6–12 reps at 70%–85% 1RM to muscle failure, two to three times per week for 8–12 wk may be the minimal training dose required to increase 1RM strength (23). In this regard, an even lower training dose, as induced by VL0, which consisted of three sets of one rep at 70%–85% 1RM twice a week for 8 wk produced suboptimal yet significant increases in 1RM strength. Likewise, a previous study showed increased gains in 1RM strength derived from VL20 versus VL40 (18.0% vs 13.4%, respectively) (15). In addition, the VL20 group showed significant gains in high-velocity actions (i.e., AV > 1), whereas VL40 did not (6.2% vs 1.0%, respectively) (15). However, previous studies on this topic only analyzed two VL thresholds (15,16); thus, the dose–response relationship between RT-induced adaptations and the whole spectrum of relative VLs (i.e., from no VL (0%) to muscle failure (~40%)) remained to be investigated. Moreover, the VL10 and VL20 groups experienced earlier significant improvements in 1RM strength (noted at sessions 2 and 3, respectively) compared with the VL0 (noted at session 6) and VL40 (noted at the end of the RT program) groups (c.f. Fig. 4), which might be of particular relevance in sport disciplines that require the maintenance of a high strength performance level throughout the entire season.

Although it has been traditionally suggested that a high number of repetitions per set during RT should be performed

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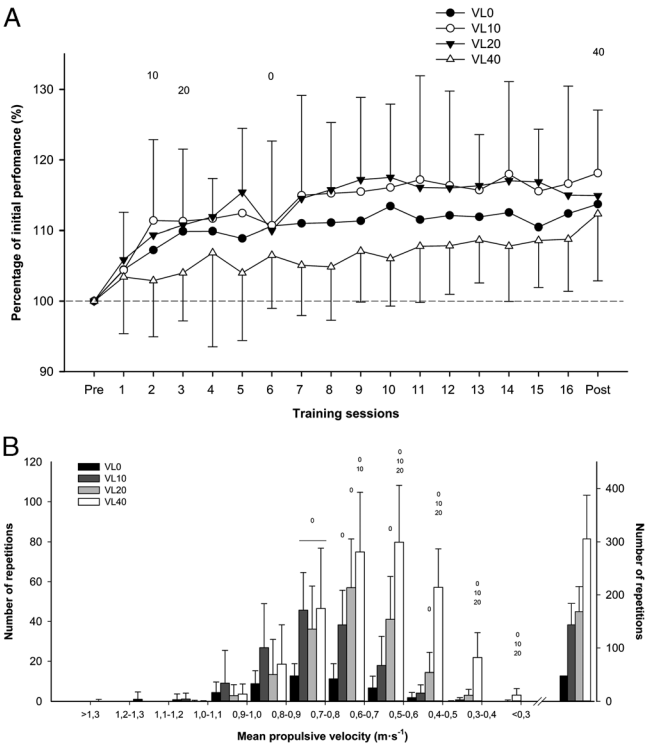

FIGURE 4—A, Evolution of the 1RM strength in the squat exercise in each training session expressed as a percentage of the initial pretraining level for each experimental group. 0, 10, 20, or 40 indicate the session from which the respective group attained significant improvements ($P < 0.05$) in 1RM strength compared with their pretraining values. B, Number of repetitions performed in each velocity range and total number of repetition completed by each training group. Between-group significant differences ($P < 0.05$): 0 for VL0, 10 for VL10, and 20 for VL20. Data are mean ± SD, $N = 55$. VL0: group that trained with a mean VL of 0% in each set ($n = 14$); VL10: group that trained with a mean VL of 10% in each set ($n = 14$); VL20: group that trained with a mean VL of 20% in each set ($n = 13$); VL40: group that trained with a mean VL of 40% in each set ($n = 14$).

to maximize gains in local muscular endurance (3,24), the four training groups all significantly improved with regard to the maximal number of repetitions achieved during the fatigue test (VL0, 70.9%; VL10, 100.2%; VL20, 103.8%; VL40, 96.3%), despite VL40 performing higher total volumes. Previous research showed greater gains in localized muscular endurance (total repetitions with 75% of 1RM) in bench press after an 11-wk training program to failure compared with nonfailure training, although there were no differences in the squat exercise (25), which suggests that the effects of level of effort during the set on endurance performance may be exercise dependent. Accordingly, our findings suggest that muscular endurance gains in SQ exercise, assessed by completing as many repetitions as possible with 70% of 1RM until the MPV fell below 0.50 $m \cdot s^{-1}$, do not seem to depend exclusively on the training volume accumulated.

In contrast, it was under moderate-to-high VL values (i.e., VL20 and VL40) that significant positive adaptations in V_{LA} muscle CSA were provoked. A previous study showed that both VL20 and VL40 increased quadriceps volume, but VL40 training elicited greater muscle hypertrophy than did VL20 (15). It has been reported that protocols with higher levels of effort during the set (i.e., higher VL) induce higher mechanical and metabolic stress (10), along with greater hormonal responses and muscle damage (11,12), which have been postulated as precursors of hypertrophic adaptations (26). It is therefore conceivable that the hypertrophic response is influenced by the level of effort attained during the set, and in this sense, both 0% and 10% of VL seemed to be insufficient to induce such adaptations. Moreover, the increases in muscle CSA in the VL20 and VL40 groups were accompanied by significant increases in FL, whereas the PA only increased significantly in the VL10 group. A further analysis in our participants showed a small but significant relationship ($r = 0.28$; $P = 0.038$) between the combined changes in FL and PA and those reported in muscle CSA. Importantly, muscle CSA and muscle architecture were measured at different muscle sites, which coupled with possible differences in regional muscle adaptations induced by RT (27,28), may explain discrepancies between the sets of measures. Our findings suggest that unlike changes in muscle function, structural muscle adaptations rely on higher levels of fatigue and training volume, as supported by previous investigations (29,30).

To our knowledge, this is the first study to assess longitudinal adaptations of contractile properties to systematic RT using simple, noninvasive and selective TMG. TMG has been validated for the assessment of *in vivo* passive muscle contractile properties in response to single-twitch stimulation (31). Based on our findings, it seems that the tested VL thresholds induced different adaptive responses in contractile properties, because only VL0 and VL10 showed significant decreases in V_{ME}-Dm, whereas VL40 showed a longer V_{LA}-Td at the end of the RT program. In this regard, plyometric training has previously been shown to produce decreases in Dm in both young active men (32) and seniors (33), which has been interpreted as an increase in muscle stiffness (21,34). Moreover, Pisot et al. (34) reported an inverse relationship ($r = -0.71$) between change in Dm and a decrease in muscle thickness after 35 d of bed rest. On the other hand, Td is recognized as a measure of muscle responsiveness (21), and has been found to be lower in power athletes compared with endurance athletes and to be negatively associated with CMJ height (35). Furthermore, a previous research found a positive relationship between the proportion of myosin heavy chain I and Td ($r = 0.61$) (36). Therefore, higher Td values may indicate negative neuromuscular adaptations. Likewise, VL40 showed significant impairment in the early RFD (RFD_{0-50}), although no significant changes were observed for RMS during MVIC for any group. Resistance exercise with high VL thresholds (i.e., VL40) is characterized by accumulated higher training volumes accompanied by large impairments in force production (10,11), which induce a reduction of IIX fiber-type percentage, whereas moderate VL thresholds (i.e., VL20) preserve this pool (15). In this regard, a positive relationship ($r = 0.61$) has been shown between the changes in early RFD (<100 ms) and changes in the area percentage of IIX fiber type (37). Thus, it is suggested that moderate VL thresholds (≤20%) should be established as maximal, in order to avoid the negative neuromuscular adaptations observed in VL40.

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Finally, of more functional relevance for coaches and practitioners were our results on RT-induced adaptations in CMJ and sprint performance. With regard to sprint performance, no significant changes were observed in any group for sprint-related variables, except for the flying phase (from 10 to 20 m), which improved in the VL10 group. By contrast, all groups significantly improved CMJ height; however, the lower VL threshold, the higher training efficiency, because all groups obtained similar CMJ improvements but accumulated substantially different total training volume. The lower VL thresholds seemed to be more efficient, and they do not evoke the high levels of discomfort and fatigue typically experienced in training routines characterized by high VL thresholds (11).

Several study limitations should be considered when interpreting our results. First, there was no control of energy or protein intake during the study, which can influence RT-induced adaptations. Second, the study was carried out in the SQ exercise, and it should be noted that VL in a multijoint exercise such as the SQ may involve peripheral fatigue in one or more muscles at different joints. Thus, the findings may not be generalizable to single joint exercise, where fatigue necessarily only manifests in muscles of an isolated joint. Moreover, the findings may not be generalizable to upper limb exercises as well. Third, no control group or control period without training before the start of the training intervention (e.g., −2 wk and week 0) was included, and as a consequence, the influence of environmental variables cannot be ruled out. Fourth, RFD findings should be interpreted with caution because the reliability values observed in RFD measurements were limited (CV from 12.8% to 24.3%). Lastly, various fitness tests were performed on day 2 (1: running sprint; 2: vertical jump; 3: MVIC in SQ; 4: progressive loading test in SQ; and 5: fatigue test in SQ), which may have induced fatigue in the subjects. Although all groups would have been affected similarly by this fact, it still raises the possibility that the tests themselves may not be as accurate in determining the intended outcomes as a result.

CONCLUSIONS

The main findings of the current study were that higher VL thresholds (i.e., VL20 and VL40) maximized hypertrophic adaptations, although an excessive level of effort during the set (i.e., VL40) may also induce negative neuromuscular adaptations. Moreover, no differences between groups were observed for RT-induced gains in squat and jump performance, despite the considerable differences in the total volume accumulated by each group. Therefore, setting a moderate VL threshold (i.e., VL20) during RT might serve to avoid performing fatiguing repetitions, which does not elicit additional benefits in terms of muscle strength and might evoke negative neuromuscular adaptations.

PRACTICAL APPLICATIONS

The results of the present study contribute to improving our knowledge about RT design and prescription. By monitoring repetition velocity, a VL threshold should be chosen beforehand, depending on the specific training goal being pursued. Moderate-to-high VL thresholds (i.e., VL20 and VL40) should be used with the aim of maximizing muscle hypertrophy, although an excessive VL during the set (i.e., VL40) should be avoided to prevent negative neuromuscular adaptations. Furthermore, moderate VL thresholds (i.e., VL10 and VL20) should be chosen to maximize athletic performance because VL0 seemed to induce levels of fatigue that were too low to maximize adaptations, whereas VL40 did not produce further strength gains compared with VL10 and VL20. Therefore, moderate VL thresholds should be chosen to maximize strength adaptations and to prevent negative neuromuscular adaptations.

The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine. This study has no conflicts of interest to declare. This study was not funded.

REFERENCES

1. Suchomel TJ, Nimphius S, Bellon CR, Stone MH. The importance of muscular strength: training considerations. *Sports Med*. 2018;48(4):765–85.
2. Schoenfeld BJ, Ogborn D, Krieger JW. Dose–response relationship between weekly resistance training volume and increases in muscle mass: a systematic review and meta-analysis. *J Sports Sci*. 2017;35(11):1073–82.
3. Kraemer WJ, Ratamess NA. Fundamentals of resistance training: progression and exercise prescription. *Med Sci Sports Exerc*. 2004;36(4):674–88.
4. Gonzalez-Badillo JJ, Gorostiaga EM, Arellano R, Izquierdo M. Moderate resistance training volume produces more favorable strength gains than high or low volumes during a short-term training cycle. *J Strength Cond Res*. 2005;19(3):689–97.
5. Gonzalez-Badillo JJ, Izquierdo M, Gorostiaga EM. Moderate volume of high relative training intensity produces greater strength gains compared with low and high volumes in competitive weightlifters. *J Strength Cond Res*. 2006;20(1):73–81.
6. Busso T. Variable dose–response relationship between exercise training and performance. *Med Sci Sports Exerc*. 2003;35(7):1188–95.
7. Barbalho M, Coswig VS, Steele J, Fisher JP, Paoli A, Gentil P. Evidence for an upper threshold for resistance training volume in trained women. *Med Sci Sports Exerc*. 2019;51(3):515–22.
8. Ostrowski KJ, Wilson GJ, Weatherby R, Murphy PW, Lyttle AD. The effect of weight training volume on hormonal output and muscular size and function. *J Strength Cond Res*. 1997;11(3):148–54.
9. Ralston GW, Kilgore L, Wyatt FB, Baker JS. The effect of weekly set volume on strength gain: a meta-analysis. *Sports Med*. 2017;47(12):2585–601.
10. Sanchez-Medina L, Gonzalez-Badillo JJ. Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Med Sci Sports Exerc*. 2011;43(9):1725–34.
11. Pareja-Blanco F, Rodriguez-Rosell D, Aagaard P, et al. Time course of recovery from resistance exercise with different set configurations [published online ahead of print July 20, 2018]. *J Strength Cond Res*. doi: 10.1519/JSC.0000000000002756.
12. Moran-Navarro R, Perez CE, Mora-Rodriguez R, et al. Time course of recovery following resistance training leading or not to failure. *Eur J Appl Physiol*. 2017;117(12):2387–99.
13. Richens B, Cleather DJ. The relationship between the number of repetitions performed at given intensities is different in endurance and strength trained athletes. *Biol Sport*. 2014;31(2):157–61.
14. Rodriguez-Rosell D, Yanez-Garcia JM, Sanchez-Medina L, Mora-Custodio R, Gonzalez-Badillo JJ. Relationship between velocity loss

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and repetitions in reserve in the bench press and back squat exercises [published online ahead of print April 25, 2019]. *J Strength Cond Res*. doi: 10.1519/JSC.0000000000002881.
15. Pareja-Blanco F, Rodriguez-Rosell D, Sanchez-Medina L, et al. Effects of velocity loss during resistance training on athletic performance, strength gains and muscle adaptations. *Scand J Med Sci Sports*. 2017;27(7):724–35.
16. Pareja-Blanco F, Sanchez-Medina L, Suarez-Arrones L, Gonzalez-Badillo JJ. Effects of velocity loss during resistance training on performance in professional soccer players. *Int J Sports Physiol Perform*. 2017;12(4):512–9.
17. Csapo R, Alegre LM, Baron R. Time kinetics of acute changes in muscle architecture in response to resistance exercise. *J Sci Med Sport*. 2011;14(3):270–4.
18. Alegre LM, Ferri-Morales A, Rodriguez-Casares R, Aguado X. Effects of isometric training on the knee extensor moment-angle relationship and vastus lateralis muscle architecture. *Eur J Appl Physiol*. 2014;114(11):2437–46.
19. Hermens HJ, Freriks B, Disselhorst-Klug C, Rau G. Development of recommendations for SEMG sensors and sensor placement procedures. *J Electromyogr Kinesiol*. 2000;10(5):361–74.
20. Krizaj D, Simunic B, Zagar T. Short-term repeatability of parameters extracted from radial displacement of muscle belly. *J Electromyogr Kinesiol*. 2008;18(4):645–51.
21. Macgregor LJ, Hunter AM, Orizio C, Fairweather MM, Ditroilo M. Assessment of skeletal muscle contractile properties by radial displacement: the case for tensiomyography. *Sports Med*. 2018;48(7):1607–20.
22. Sanchez-Medina L, Perez CE, Gonzalez-Badillo JJ. Importance of the propulsive phase in strength assessment. *Int J Sports Med*. 2010; 31(2):123–9.
23. Androulakis-Korakakis P, Fisher JP, Steele J. The minimum effective training dose required to increase 1RM strength in resistance-trained men: a systematic review and meta-analysis. *Sports Med*. 2020;50(4): 751–65.
24. Bird SP, Tarpenning KM, Marino FE. Designing resistance training programmes to enhance muscular fitness: a review of the acute programme variables. *Sports Med*. 2005;35(10):841–51.
25. Izquierdo M, Ibanez J, Gonzalez-Badillo JJ, et al. Differential effects of strength training leading to failure versus not to failure on hormonal responses, strength, and muscle power gains. *J Appl Physiol*. 2006;100(5):1647–56.
26. Schoenfeld BJ. The mechanisms of muscle hypertrophy and their application to resistance training. *J Strength Cond Res*. 2010;24(10): 2857–72.
27. Earp JE, Newton RU, Cormie P, Blazevich AJ. Inhomogeneous quadriceps femoris hypertrophy in response to strength and power training. *Med Sci Sports Exerc*. 2015;47(11):2389–97.
28. Mangine GT, Redd MJ, Gonzalez AM, et al. Resistance training does not induce uniform adaptations to quadriceps. *PLoS One*. 2018; 13(8):e0198304.
29. Mitchell CJ, Churchward-Venne TA, West DW, et al. Resistance exercise load does not determine training-mediated hypertrophic gains in young men. *J Appl Physiol (1985)*. 2012;113(1):71–7.
30. Schoenfeld BJ, Contreras B, Krieger J, et al. Resistance training volume enhances muscle hypertrophy but not strength in trained men. *Med Sci Sports Exerc*. 2019;51(1):94–103.
31. Zagar T, Krizaj D. Validation of an accelerometer for determination of muscle belly radial displacement. *Med Biol Eng Comput*. 2005; 43(1):78–84.
32. Zubac D, Simunic B. Skeletal muscle contraction time and tone decrease after 8 weeks of plyometric training. *J Strength Cond Res*. 2017;31(6):1610–9.
33. Zubac D, Paravlic A, Koren K, Felicita U, Simunic B. Plyometric exercise improves jumping performance and skeletal muscle contractile properties in seniors. *J Musculoskelet Neuronal Interact*. 2019;19(1): 38–49.
34. Pisot R, Narici MV, Simunic B, et al. Whole muscle contractile parameters and thickness loss during 35-day bed rest. *Eur J Appl Physiol*. 2008;104(2):409–14.
35. Loturco I, Gil S, Laurino CF, et al. Differences in muscle mechanical properties between elite power and endurance athletes: a comparative study. *J Strength Cond Res*. 2015;29(6):1723–8.
36. Simunic B, Degens H, Rittweger J, Narici M, Mekjavic IB, Pisot R. Noninvasive estimation of myosin heavy chain composition in human skeletal muscle. *Med Sci Sports Exerc*. 2011;43(9):1619–25.
37. Andersen LL, Andersen JL, Zebis MK, Aagaard P. Early and late rate of force development: differential adaptive responses to resistance training? *Scand J Med Sci Sports*. 2010;20(1):e162–9.

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